

Ioannis (Giannis) Georgiou

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Education

University of Manitoba

Bachelor of Science in Computer Engineering

Winnipeg, MB

Expected May 2027

Relevant Coursework: Real-time Embedded Systems, Modern Computer Architecture, Microprocessor Architecture and Interfacing, Data Structures and Algorithms, Object-Oriented Design & Software Architecture

Engineering Experience

University of Manitoba Space Applications and Technology Society (UMSATS)

Avionics Subsystem Member

Winnipeg, MB

Aug 2024 – Jan 2026

- Conducted research, designed and tested **flight computer systems** for student-researched and developed (SRAD) rockets.
- Designed PCBs using **KiCad**, integrating microcontrollers, sensors, and communication interfaces.
- Implemented and validated embedded firmware for microcontroller-based flight systems, supporting hardware/software integration.

University of Manitoba Space Technology and Advanced Research Lab (STARLab)

Undergraduate Student Researcher

Winnipeg, MB

May 2024 – Aug 2024

- Co-authored a paper published at **SpaceOps 2025** on an **EKF** framework for real-time CubeSat attitude determination with gyroscope and magnetometer bias estimation.
- Configured and verified a 12-camera **VICON** motion capture system as experimental ground truth, confirming angular tracking accuracy within **0.07%** at 15 RPM.
- Utilized **MATLAB** to phase-align **BNO055** IMU data with VICON reference using cross-correlation and rotation matrix transformations.

Projects

Wireless Visual Command System for Deaf Curlers

ESP32 | Onshape | C

- Developed a wireless communication system using **ESP32** and the **ESP-NOW** protocol to assist deaf athletes in receiving real-time curling commands
- Implemented **C** firmware for a multiplexed reversing-radar display to display distinct visual curling instructions like "Curl" or "Line".
- Designed 3D-printed ergonomic cases for a handheld transmitter controller and wearable hat-mounted receiver device using **Onshape**.

Automotive Airbag Simulator

STM32 | FreeRTOS | C

- Developed a real-time simulation of an automotive airbag system in **C** using an **STM32F407** board.
- Designed preemptive task scheduling and inter-process communication in **FreeRTOS**, utilizing hardware interrupts, binary semaphores, and shared memory.
- Implemented airbag deployment logic with accurate GPIO signaling and UART logging for timing validation.

Real-Time Gesture Recognition System

TensorFlow Lite Micro | Python | C++

- Trained and int8-quantized a neural network in **Python**, deploying it for real-time inference on a **Raspberry Pi Pico** with no host computer required.
- Ported feature extraction (mean, standard deviation, min, max per sensor axis) and int8 quantization logic from **Python** to **C++**.
- Built an **I2C** sensor driver, windowing buffer, and inference loop in **C++**, verifying each stage independently before integration.

Technical Skills

Programming & HDL: Java, Python, C, ARM Assembly, MATLAB, Verilog HDL

Embedded Systems: STM32, ESP32, FreeRTOS, UART/SPI/I2C

Tools & Design: KiCad, EasyEDA, Git, LaTeX, Onshape